

# EXPERIMENT 14

## AIM

Calculate MD5, SHA-1 and SHA-256 for the same evidence file and compare them with recorded values.

## ALGORITHM

1. Specify the evidence file.
2. Create MD5, SHA-1 and SHA-256 hash objects.
3. Open the file in binary mode.
4. Read the file in chunks.
5. Update all hash objects with each chunk.
6. Display the final hexadecimal digests.
7. Stop.

## CODE

```
import hashlib

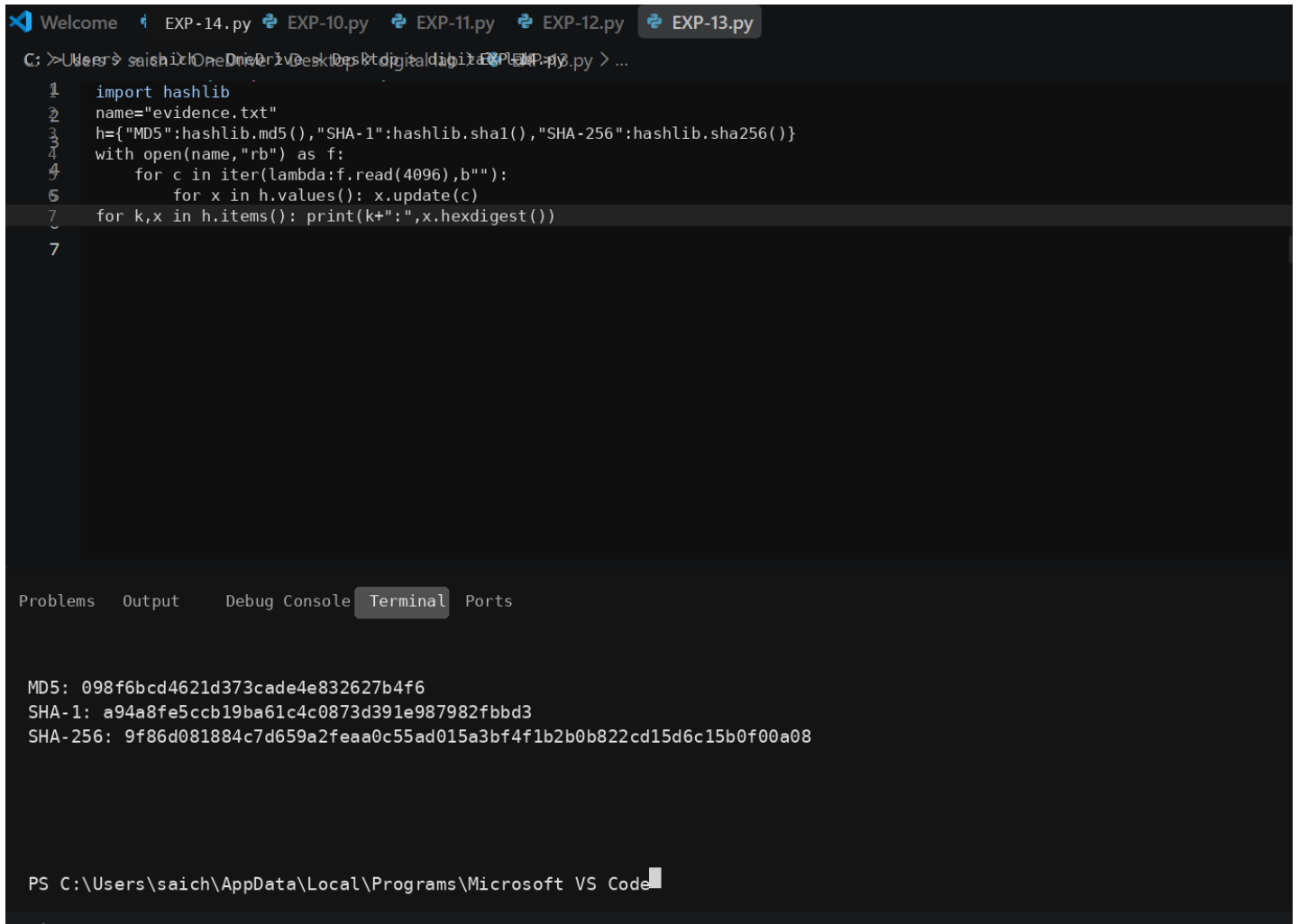
filename = "evidence.txt"

hashers = {
    "MD5": hashlib.md5(),
    "SHA-1": hashlib.shal(),
    "SHA-256": hashlib.sha256()
}

with open(filename, "rb") as file:
    for chunk in iter(lambda: file.read(4096), b''):
        for hasher in hashers.values():
            hasher.update(chunk)

for name, hasher in hashers.items():
    print(f"{name}: {hasher.hexdigest()}")
```

## OUTPUT



The image shows a Visual Studio Code editor window with a file explorer on the left and a code editor in the center. The file explorer shows a folder named 'EXP-13.py' with a subfolder 'EXP-13.py'. The code editor displays a Python script named 'EXP-13.py' with the following code:

```
1 import hashlib
2 name="evidence.txt"
3 h={"MD5":hashlib.md5(),"SHA-1":hashlib.sha1(),"SHA-256":hashlib.sha256()}
4 with open(name,"rb") as f:
5     for c in iter(lambda:f.read(4096),b""):
6         for x in h.values(): x.update(c)
7     for k,x in h.items(): print(k+":",x.hexdigest())
```

The terminal output shows the following hashes:

```
MD5: 098f6bcd4621d373cade4e832627b4f6
SHA-1: a94a8fe5ccb19ba61c4c0873d391e987982fbbd3
SHA-256: 9f86d081884c7d659a2feaa0c55ad015a3bf4f1b2b0b822cd15d6c15b0f00a08
```

The terminal status bar at the bottom indicates the file path: PS C:\Users\saich\AppData\Local\Programs\Microsoft VS Code

## RESULT

The utility calculates three cryptographic hashes for forensic evidence verification.